

Ch 2 File Systems in detail

- DBMS is built on top of a file system
- a file is a collection of bytes stored on secondary storage and managed by OS

File system

↳ component of OS responsible for organizing and managing files

- create / delete / rename files
- read / write bytes
- manage directories / permissions at file level

Flow

Application → read() → OS → file system → disk

application itself parses CSV, search and update records validate data, OS does none of this

Text files

PROS	CONS
• easy debugging • portable • platform independent	• no schema • no indexing • slow process

CSV files

(Comma separated values)

PROS	CONS
• easy import/export • excel supported	• no indexing • no constraint • no relation • no transaction

JSON files

(excellent to represent semi structured data)

PROS	CONS
• flexible schema • easy to exchange over API's • supports nested objects	• no transaction • no concurrency • searching needs scanning • large dataset is inefficient

File system vs DBMS

Feature	File system	DBMS
stores bytes	✓	✓
query language	✗	✓
indexing	✗	✓
transaction	✗	✓

Concurrency control	✗	✓
recovery	✗	✓
integrity	✗	✓
query optimization	✗	✓
security	✗	✓

Sequential access vs random access

Sequential access	random access
• $O(n)$ • looks for data in a sequential manner, i.e checks all the record • good for logs, large sequential reads, streaming etc	• $O(1)$ • Jump directly to required location using index/offset • takes use of B+ trees and hash indices